(KTXFI1EBNF) Physics I. Lecture

Dr. Gábor FACSKÓ, PhD
Assistant Professor / Senior Research Fellow
facsko.gabor@uni-obuda.hu

Óbuda University, Faculty of Electrical Engineering, 1084 Budapest, Tavaszmező u. 17.
Wigner Research Centre for Physics, Department of Space Physics and Space Technology, 1121 Budapest, Konkoly-Thege Miklós út 29–33.
https://wigner.hu/~facsko.gabor

April 20, 2026

Current Updates and Announcements

- April 20, April 27, May 4, and go...
- There will be one occasion without a lecture.
- Homework No. 2 is now available soon. The deadline is May 4, 2026 at midnight.
- Slides on Moodle: better oriented lecture, and practice with solution.

Where are we? I

1. **Mechanics I.:** Newton's laws, equation of motion (review). Momentum. General concept of mechanical work, work theorem. Forms of mechanical energy.
2. **Mechanics II.:** Weight force, gravitational force, gravitational field. Work of the gravitational force. Conservative force field. Dissipative force field. Mechanical power.
3. **Mechanics of Rigid Bodies I.** Concept of a rigid body. Forces acting on a rigid body. Motion of a rigid body. Concept of the centre of mass. Location of the centre of mass. Torque. Angular momentum. Energy of a rotating rigid body, angular momentum of a rotating rigid body, Steiner's theorem.
4. **Mechanics of Rigid Bodies II.** Principal theorems of rigid body motion; conservation laws. Collisions.
5. **Oscillations I.** Dynamic condition of harmonic oscillatory motion. Differential equation of harmonic undamped oscillatory motion.

Where are we? II

6. **Oscillations II.** Superposition of oscillations. Decomposition of oscillations into components. Damped oscillations. Forced oscillations, resonance. Coupled oscillations.
7. **Wave phenomena:** Reflection of waves, refraction of waves, diffraction of waves. Huygens–Fresnel principle. Interference of waves. Polarization of waves.
8. **Elements of acoustics:** Basics of acoustics. Fundamental concepts. Doppler effect. Sound intensity, loudness according to Weber–Fechner's law.
9. **Thermodynamics I:** Temperature scales, thermal expansion, thermometers. State variables, ideal gas, state changes, ideal gas equation of state, Boyle's law, Gay-Lussac's laws, universal (combined) gas law, heat, specific heat, molar heat, heat capacity.
10. **Thermodynamics II:** Expansion work, internal energy, the first law of thermodynamics, forms of the first law in different state changes.

Where are we? III

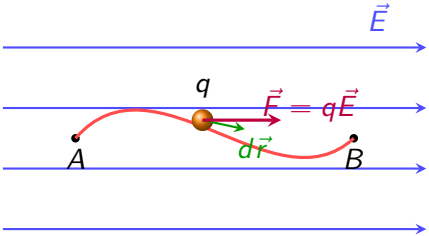
11. **Thermodynamics III:** Cyclic processes, Carnot cycle, Carnot's theorem, reduced heat, entropy, the second law of thermodynamics.
12. **Thermodynamics IV:** Kinetic theory of gases, molecular thermodynamics. Distributions, distribution functions. Elementary statistics.
13. **Electrostatics I.:** Electric charge: Fundamental electrical phenomena, electric charge and matter, Coulomb's law (force between two point charges, unit of electric charge, permittivity of vacuum).
14. **Electrostatics II.:** The electric field: The electrostatic field (electric field strength, electric field lines). Determination of electric field strength. Gauss's law (flux of the electric field). Electric field of a point charge.

Where are we? IV

15. **Electrostatics III.:** *Electrostatic potential: Work of the electric field (work done in an electric field). Electric potential (electrostatic potential). Electric potential difference, voltage. Electric potential of charge systems at small and large distances. Electrostatic energy.*
16. **Elements of physical or wave optics.** Branches of optics. Light interference, light diffraction, diffraction through a slit, optical grating, light polarization. Light as an electromagnetic wave: Poynting vector. Optical instruments.

Electrostatic Potential I

- **Work Done in an Electric Field**



- When a charge q moves in an electric field $\vec{E}$, the electric force is
$$\vec{F} = q\vec{E}.$$
- The work done by the electric field when the charge moves from point A to point B is
$$W_{A\to B} = \int_A^B \vec{F}\cdot d\vec{r} = q\int_A^B \vec{E}\cdot d\vec{r}.$$

- The work depends on the electric field and the path.
- In electrostatics, the field is conservative.
- Therefore, the work depends only on the initial and final points.

Electrostatic Potential II

- **Conservative Nature of the Electrostatic Field:** For an electrostatic field,

$$\oint \vec{E} \cdot d\vec{r} = 0.$$

- Therefore,
  - The total work along any closed path is zero.
  - We can define a scalar quantity called **electric potential**.
  - It is often easier to work with potential than with force.

$$W_{A \to B} = -\Delta U = -(U_B - U_A),$$

  where U is the electric potential energy.

- **Remember: Gravity!!!**

Electrostatic Potential III

- **Electric Potential and Voltage:** The **electric potential** (electrostatic potential) is the potential energy per unit charge:

$$V = \frac{U}{q}.$$

- The potential difference between points A and B is

$$V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{r}.$$

  - Unit of potential: volt (V)
  - $1\,\mathrm{V} = 1\,\mathrm{J/C}$
  - Potential is a scalar quantity

Electrostatic Potential IV

- **Potential Difference and Voltage:** The **voltage** between two points is their potential difference:

$$U_{AB} = V_A - V_B.$$

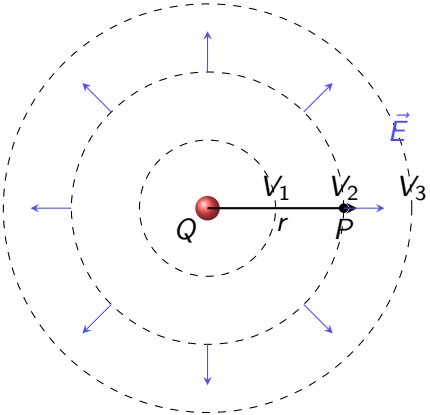
- Important comments:
  - If a positive charge moves in the direction of the electric field, the potential decreases.
  - The electric field points from higher potential to lower potential.
  - Relation between field and potential:

$$\vec{E} = -\nabla V.$$

- **Remark:** $\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$, gradient, divergence, rotation.

Electrostatic Potential V

- **Potential of a Point Charge**



- For a point charge Q, the electric potential at distance r is
$$V(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r},$$
if we choose $V(\infty) = 0$.
  - If $Q > 0$, then $V > 0$
  - If $Q < 0$, then $V < 0$
  - Potential decreases as $1/r$
- The electric field of a point charge is
$$E(r) = \frac{1}{4\pi\varepsilon_0}\frac{|Q|}{r^2}.$$

Electrostatic Potential VI

- **Potential of Several Charges:** For a system of point charges, potentials add algebraically:

$$V(\vec{r}) = \sum_i \frac{1}{4\pi\varepsilon_0} \frac{q_i}{r_i},$$

  where r_i is the distance from charge q_i to the observation point.
  - Potential obeys the superposition principle.
  - Since potential is a scalar, addition is simpler than for electric field.
- For a continuous charge distribution:

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{dq}{r}.$$

Electrostatic Potential VII

- **Potential at small distances:**
  - The nearest charges have the strongest effect.
  - The detailed geometry of the charge system is important.
- **Potential at large distances:**
  - A finite charge system often behaves like a single effective charge.
  - If the total charge is Q_{tot}, then approximately

$$V(r) \approx \frac{1}{4\pi\varepsilon_0}\frac{Q_{\text{tot}}}{r}.$$

- If the total charge is zero, the leading term may be a dipole term, which decreases faster with distance.

Electrostatic Potential VIII

- **Example: Electric Dipole:** An electric dipole consists of charges $+q$ and $-q$ separated by a small distance.
  - Close to the dipole, the potential depends strongly on position.
  - Far from the dipole, the potential is much smaller than for a single charge.
- For large distances, the dipole potential behaves approximately as

$$V(r,\theta) \sim \frac{\cos\theta}{r^2}.$$

- Therefore,
  - point charge: $V \sim 1/r$
  - dipole: $V \sim 1/r^2$

Electrostatic Potential Energy I

- If a charge q is placed at a point where the potential is V, then its **potential energy** is
$$U = qV.$$
- For two point charges:
$$U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}.$$
- Interpretation:
  - $U > 0$: energy must be supplied to assemble the system
  - $U < 0$: energy is released when the system is assembled

Electrostatic Potential Energy II

- **Electrostatic Energy of a Charge System:** For many point charges, the total electrostatic energy is

$$U = \frac{1}{2} \sum_i q_i V(\vec{r}_i),$$

where $V(\vec{r}_i)$ is the potential produced by all other charges at the position of q_i.

- Equivalent form:

$$U = \frac{1}{4\pi\varepsilon_0} \sum_{i<j} \frac{q_i q_j}{r_{ij}}.$$

  - The factor $\frac{1}{2}$ avoids double counting.
  - This energy is stored in the electric configuration.

Electrostatic Potential Energy III

- **Energy Stored in the Electric Field:** Electrostatic energy can also be described as energy stored in the field itself.
- The energy density is
$$u = \frac{1}{2}\varepsilon_0 E^2.$$
- So the total energy is
$$U = \int u \, d\tau = \frac{1}{2}\varepsilon_0 \int E^2 \, d\tau.$$
- This form is especially useful for continuous charge distributions and capacitors.

The End

Thank you for your attention!